

GoFlockYourself

Passive Flock Safety Camera Detector
for the Cheap Yellow Display (ESP32-2432S028R)

Technical Documentation

Version 1.0.0

12 August 2026

By @OffGridPete

A production-oriented firmware and field guide for passively detecting Flock Safety and related automatic license plate reader (ALPR) infrastructure using WiFi promiscuous mode, OUI fingerprinting, wildcard probe requests, and optional Bluetooth Low Energy scanning — all on a low-cost ESP32 touchscreen board.

Table of Contents

- 1. Background & Research
- 2. Limitations
- 3. Hardware Requirements
- 4. Getting Started / How to Use
- 5. User Interface & Navigation
- 6. Detection Methods
- 7. Alerts & Sensitivity
- 8. Logging & microSD Card
- 9. Best Practices & Tips
- 10. Credits & References
- Appendix A — Confidence Scoring
- Appendix B — Serial Diagnostics
- Appendix C — File Map

1. Background & Research

1.1 What Are Flock Safety Cameras?

Flock Safety manufactures networked automatic license plate reader (ALPR) systems deployed by municipalities, law enforcement agencies, homeowners associations, and private security clients primarily across the United States (and to a lesser extent Canada). Units are commonly mounted on utility poles, streetlights, or dedicated stands near roadways. Product families observed in the field include Falcon and Sparrow style pole cameras, as well as related surveillance hardware from adjacent vendors in the broader ALPR ecosystem.

These systems capture vehicle images and plate data, often correlated with time, location, and vehicle attributes, and upload that data to vendor cloud services. Because deployments can be dense, opaque, and operated with limited public notice, journalists, civil liberties advocates, municipal watchdogs, and privacy-conscious residents have developed independent methods to map where such infrastructure exists.

1.2 Why Detect Them?

Independent detection does not jam, disable, or interfere with cameras. It enables:

- **Transparency** — Cross-check public records and vendor maps against RF signatures observed on the ground.
- **Privacy research** — Document the density and placement of always-on ALPR infrastructure in a community.
- **Field auditing** — Verify whether a reported camera location is active and radiating 2.4 GHz management traffic.
- **Education** — Teach how commodity WiFi/BLE receivers can reveal device classes without network association.

Legal and ethical framing

GoFlockYourself is a **receive-only** tool. It does not transmit deauthentication frames, does not associate with camera networks, and does not attempt to access camera data. Users remain responsible for complying with local law regarding radio reception and surveillance mapping.

1.3 How Passive Detection Became Possible

ALPR cameras that use 2.4 GHz WiFi for configuration, management, or uplink discovery emit standard IEEE 802.11 frames. Even when they do not host a visible access point, they may act as stations (STAs) searching for a hidden management SSID. Those emissions are public over the air and can be observed with a WiFi radio in **promiscuous (monitor) mode** — the primary technique used by GoFlockYourself.

Promiscuous WiFi mode

In promiscuous mode, the ESP32 radio delivers management and data frames from the air to firmware without joining a network. The detector hops across 2.4 GHz channels (typically 1, 6, and 11, or the full 1–13 set) and inspects each frame's addresses and information elements (IEs).

OUI fingerprinting

Every MAC address begins with a 24-bit Organizationally Unique Identifier (OUI). Researchers compiled prefixes repeatedly associated with Flock Safety hardware, including modules from suppliers such as LiteOn and related silicon vendors. GoFlockYourself ships a compiled table of approximately **41 OUI prefixes**, including the IEEE MA-L block **B4:1E:52** registered to Flock Safety (9 May 2024).

addr2 (transmitter) and addr1 (receiver) matching

In 802.11 headers, **addr2** is typically the transmitter address and **addr1** the receiver. Matching a known OUI on addr2 catches cameras that are actively transmitting. Matching on addr1 catches nearby access points or peers that are talking to a Flock MAC — a technique highlighted in NitekryDPaul's research even when the camera itself is quiet on that hop.

Wildcard probe requests

A probe request with an empty SSID Information Element (tag 0, length 0) is a **wildcard probe**: the station is asking any AP to respond. Field research (notably DeFlockJoplin / Michael) observed Flock cameras emitting wildcard probes while hunting uplink networks, often on a rapid channel sequence. Combined with a known OUI on the transmitter, this is currently among the **highest-confidence** passive signatures available.

1.4 Evolution of Camera RF Behavior

Camera firmware and backhaul strategy have changed over time. Operators and researchers should treat historical detection recipes as provisional:

- **Earlier generations** often exposed more continuous WiFi AP or management beacons; some community tooling targeted those SSID patterns directly.
- **~Late 2025 reports** describe reduced reliance on always-on WiFi management APs, with devices more often in STA mode probing for hidden uplinks.
- **Cellular (LTE) backhaul** is increasingly common. Units that primarily use LTE may emit little or no 2.4 GHz WiFi for long periods.
- **BLE advertising** that once helped detect Raven/Flock-adjacent gear has been reported as less reliable after vendor updates.
- **Locally administered MACs** (U/L bit set) appear on some units; the firmware still matches exact OUI triples from research lists (for example 82:6B:F2) rather than blanket-rejecting all local-admin addresses.

GoFlockYourself therefore prioritizes **WiFi promiscuous OUI + wildcard probe** detection, treats BLE as a secondary interleaved method, and documents honesty about miss rates for cellular-first deployments.

1.5 Research Credits (Summary)

This project stands on community research rather than inventing detection from scratch. Primary credits include @NitekryDPaul (OUI corpus and addr1 technique), Michael / DeFlockJoplin (wildcard probe signature and field validation), colonelpanichacks / flock-you (promiscuous pipeline reference), GainSec (Raven BLE UUID research), and the broader DeFlock mapping community. See Section 10 for the full acknowledgments list.

2. Limitations

Read this section carefully

GoFlockYourself is a **useful field aid**, not a guaranteed detector. Absence of an alert does **not** prove absence of a camera. Presence of an alert is a probabilistic signal that must be interpreted with context (RSSI, method, location, repeatability).

2.1 Not 100% Reliable

No passive RF tool can claim perfect recall against a product line that can silence or relocate its radio stack via OTA updates. Detection depends on the camera emitting something observable on 2.4 GHz WiFi or BLE during the brief windows the receiver is listening on the correct channel.

2.2 Cellular-First and Quiet Radios

Newer installations frequently prefer **LTE / cellular backhaul**. Those units may:

- Emit little or no WiFi for hours or days while operating normally.
- Only brief WiFi activity during install, recovery, or rare reconfiguration.
- Produce zero BLE advertisements of interest.

In those cases, public records, visual survey, and crowdsourced maps remain essential complements to RF detection.

2.3 Shared Manufacturer OUIs and False Positives

Many OUIs on the research list are allocated to component vendors (for example WiFi module manufacturers) rather than exclusively to Flock Safety. Devices unrelated to ALPR can share those prefixes. Wildcard-probe + OUI combinations raise confidence, but **broad OUI matching alone is noisier** and is off by default for that reason.

2.4 Duty Cycle and Sleep

Cameras spend much of their life not probing. A drive-by on a single channel plan may simply miss a quiet interval. Longer dwell, full channel plans, repeat passes, and stationary observation improve odds but never eliminate misses.

2.5 Range and Environment

Practical 2.4 GHz range for management frames depends on antenna, orientation, vehicle cabin attenuation, multipath, and interference. Expect tens of meters in favorable line-of-sight conditions and substantially less inside dense urban canyons or from a sealed metal vehicle with a poorly placed board.

2.6 Hardware Constraints

The ESP32-2432S028R has **no PSRAM** on most units, limited free GPIO, and a single 2.4 GHz radio shared between WiFi and BLE. BLE scanning therefore **time-shares** the radio (WiFi fully stops during short BLE windows). The interface remains usable, but simultaneous full-duty WiFi+BLE is not possible.

Limitation	Impact	Mitigation
LTE-only backhaul	No RF signature	Do not rely on RF alone; use maps/visuals
Shared OUIs	False positives	Prefer wildcard+OUI; disable Broad OUI

Limitation	Impact	Mitigation
Sleep / quiet periods	Missed cameras	Repeat routes; full channel plan
Short range	Miss at speed	Slower pass; careful antenna orientation
BLE vs WiFi radio share	Coverage gaps	Leave BLE off unless needed

3. Hardware Requirements

3.1 Primary Board

GoFlockYourself targets the **ESP32-2432S028R**, commonly known as the **Cheap Yellow Display (CYD)**:

Component	Specification
MCU	Espressif ESP32-WROOM-32 (dual-core, 240 MHz, WiFi + BT)
Display	2.8" ILI9341 TFT, 240 × 320 portrait
Touch	XPT2046 resistive touchscreen
Storage slot	Onboard microSD (SPI)
Status LED	Onboard RGB LED (active-low; GPIO 4 / 16 / 17)
Memory	No PSRAM on most units — firmware is static-allocation oriented
Flash	Typically 4 MB; firmware uses a custom partition table

3.2 Recommended Additions

Item	Purpose	Notes
Passive piezo buzzer	Audible hit alert	Wire + to GPIO 22 (CN1/P3), – to GND
microSD card	CSV detection log	Must be FAT32 ; see Section 8
USB power bank	Field power	Stable 5 V; avoid weak cables
Non-conductive mount	Dash / handlebar	Keep display readable; free USB port

3.3 Power

Power the CYD via its USB port from a computer for flashing, or from a quality USB power bank in the field. Brownouts can reset the ESP32 mid-scan; if the device reboots under load, try a shorter cable and a higher-capacity pack.

3.4 Optional Buzzer Wiring

Piezo + → GPIO 22 (CN1 header)
Piezo – → GND

For a louder active buzzer, drive a small NPN/MOSFET from GPIO 22 rather than powering the buzzer directly from the pin.

microSD requirement

SD card logging works **only** when a microSD card formatted as **FAT32** is inserted in the onboard slot. Without a card, the firmware still operates fully; detections appear on-screen, in history, and on the USB serial console.

4. Getting Started / How to Use

4.1 Prerequisites

- A computer with USB and the project source tree (GoFlockYourself/).
- PlatformIO Core or the PlatformIO IDE extension for VS Code.
- A data-capable USB cable (charge-only cables will power the board but not flash it).

4.2 Flashing the Firmware

From a terminal:

```
cd GoFlockYourself
pio run -e cyd -t upload
pio device monitor -b 115200
```

On macOS, the serial device is typically of the form `/dev/cu.usbserial-*`. If upload fails with “Failed to connect,” hold the board’s **BOOT** button, tap **RST**, release **BOOT**, and retry. Close any other serial monitor that may hold the port.

4.3 First Boot

After a successful flash the board should:

- Show a brief **GoFlockYourself** splash screen.
- Enter the dark-themed **Home / Status** screen with **LIVE** scanning enabled.
- Pulse the RGB LED in a soft blue scanning pattern.
- Print a boot banner and configuration summary on USB serial at 115200 baud.
- Auto-detect a FAT32 microSD card if present and enable CSV logging.

If the display remains blank, press **RST** once. Confirm serial output contains lines such as `[wifi] SCAN start` and a periodic `[hb]` heartbeat with a non-zero frame counter when near any 2.4 GHz WiFi activity.

4.4 Basic Operation

- Leave the device scanning while walking or driving through an area of interest.
- Watch the Home screen for channel hops, hit counters, and last detection details.
- On a hit: full-screen alert, optional buzzer pattern, RGB flash, history entry, serial line, and SD append (if enabled).
- Use the touch menu to stop/start scanning, toggle methods, adjust alerts, and review history.

4.5 Power Recommendations

Use case	Recommendation
Bench / flash	Computer USB port; keep serial monitor available
Walking survey	10,000 mAh+ power bank; secure the board; avoid pocket RF shielding
Vehicle	Dedicated USB outlet or power bank; mount with display visible to a passenger
Long soak	Mains USB adapter; ensure ventilation; SD logging for unattended capture

5. User Interface & Navigation

5.1 Design Overview

The UI is a custom dark theme on the 240×320 ILI9341 panel using TFT_eSPI, with resistive touch via XPT2046. The design prioritizes field readability: high contrast, large hit counters, and a prominent detection alert. Touch targets are full-width rows where possible to compensate for resistive imprecision.

5.2 Home / Status Screen

The Home screen is the primary operational view. It shows:

Element	Meaning
Title + version	Project identity (GoFlockYourself / v1.0.0)
LIVE / IDLE	Whether WiFi promiscuous scanning is active
Channel	Current 2.4 GHz hop channel
Mode	Channel plan: Primary 1/6/11, Full 1–13, or Asc 1–11
Methods	Compact flags for enabled methods (OUI, WILD, BLE, ...)
Frames	Total frames seen (proves the radio is alive)
SD:status	microSD state: ready / none / error
HITS / UNIQUE	Total alerted events vs unique MACs in history
Last detection	Method, OUI, RSSI, channel, time ago
Uptime / Heap	Runtime and free heap for health monitoring
MENU >	Touch affordance to open the main menu

Tap the **title bar** or the lower **MENU >** region to open the main menu. Status fields refresh periodically while remaining on Home so channel and counters stay live without full-screen redraws.

5.3 Touch Navigation

- Use firm, deliberate presses; resistive panels need pressure, not a hover.
- Most sub-screens provide a **< BACK** control in the header (top-right).
- If taps feel offset, calibrate raw min/max constants in `include/config.h` (TOUCH_X/Y_MIN/MAX).
- A short debounce (~220 ms) reduces double-triggers.

5.4 Main Menu

Menu item	Function
Start / Stop Scanning	Toggle promiscuous scan; LED returns to idle green when stopped
Detection Methods	Enable/disable individual RF detection paths
Alerts & Sensitivity	Audio, LED, SD logging, channel plan
View Log / History	Paged on-device history of recent detections
About / Credits	Version, author, research credits, OUI count

5.5 Detection Methods Screen

Each row is a toggle (ON/OFF). Changes take effect immediately for subsequent frames. Rows include: WiFi OUI (TX), WiFi OUI (RX), Wildcard Probe, Broad OUI, SSID Keywords, and BLE Scanning (when compiled in). Channel plan summary is shown at the bottom for context.

5.6 Alerts & Sensitivity Screen

Toggles for Audio (buzzer), RGB LED alert, and SD logging. Live card status text reports whether a microSD is ready. Channel plan buttons select Primary (1/6/11), Full (1–13), or Ascending (1–11). See Section 7 for recommended profiles.

5.7 History Screen

Shows up to **40** recent detections in RAM (most recent first), six per page. Each row lists method, full MAC, RSSI, and relative time. **Prev** appears only when a previous page exists; **Next** only when another page exists.

5.8 About Screen

Displays project name, version, author line **By @OffGridPete**, hardware target, a short capability summary, research credits, and the number of compiled OUI prefixes.

5.9 Detection Alert Screen

On a meaningful (de-duplicated) hit, a full-screen alert flashes with method name, MAC, OUI, RSSI, channel, and confidence percentage. Tap to dismiss early, or wait for the automatic timeout (~5.5 s). Audio and LED accompany the visual alert when enabled.

6. Detection Methods

Methods are evaluated inside a lean WiFi sniffer callback that only parses and enqueues work; heavy UI, SD, and serial I/O run on the main loop. Confidence scores rank methods from 0–100 for display and triage (see Appendix A).

What works best in the field today

Highest practical confidence: **Wildcard Probe + known OUI (TX)**, optionally strengthened by LiteOn-style vendor IE fingerprints. Next: **Probe Request + OUI** and **OUI TX** on management activity. **OUI RX (addr1)** is valuable when nearby APs talk to a camera. **BLE** is secondary and off by default on the CYD. Disable **Broad OUI** unless you accept higher false-positive rates.

6.1 WiFi OUI Matching — Transmitter (addr2)

How it works

For management and (when broad mode is on) data frames, the firmware extracts **addr2** (transmitter / source). The first three octets are compared against the compiled Flock-related OUI table. Multicast and all-zero / all-ones addresses are ignored. Hits are scored and enqueued with method **OUI TX** or, if the frame is a probe request, higher-tier probe methods.

Effectiveness / confidence

Medium–high when the camera is actually transmitting. Confidence is lower than wildcard-probe combinations because many module vendors share OUIs. Baseline score for pure OUI TX is moderate (~60), elevated when coupled with probe subtypes.

When it works best

- Camera is in STA mode and actively probing or exchanging management frames.
- You pass within solid 2.4 GHz range during an active radio window.
- OUI list includes the module family used by that deployment.

Limitations

- False positives from non-ALPR devices using the same module OUI.
- Silent LTE-first units may never transmit on WiFi during your pass.
- Requires correct channel hop timing to be on-channel during emission.

Best practices

- Leave OUI TX **enabled** for general surveys.
- Triage hits by RSSI and repeatability before treating them as confirmed cameras.
- Combine with Wildcard Probe for higher confidence.

6.2 WiFi OUI Matching — Receiver (addr1)

How it works

Some frames are addressed **to** a camera even when the camera is not the transmitter. The firmware matches known OUIs on **addr1** (receiver / destination) for management frames, skipping multicast destinations. This is the **NitekryDPaul addr1 technique**: nearby infrastructure talking to a Flock MAC reveals presence indirectly.

Effectiveness / confidence

Medium. Extremely useful in dense WiFi environments near a managed uplink, but more context-dependent than transmitter probes. Score is lower than probe-based methods (~50) to reflect indirect evidence.

When it works best

- Camera has a WiFi relationship with a nearby AP that is actively sending frames.
- You are close enough to hear the AP's transmissions to that MAC.

Limitations

- Does not fire if nothing addresses the camera on-air during your listen window.
- Can confuse triage if multiple devices share related prefixes.

Best practices

- Keep enabled for thorough surveys; correlate with TX/probe hits when possible.
- Use history MAC patterns and geography to discard one-off weak RX hits.

6.3 Wildcard Probe Request Detection

How it works

The sniffer identifies IEEE 802.11 **Probe Request** management frames (type 0, subtype 4), walks Information Elements after the MAC header, and checks for SSID IE (tag 0) with **length 0**. When the transmitter OUI is also on the research list, the event is classified as **WILD PROBE** — the top-tier signature. Related probe paths also detect directed probes from known OUIs and optional LiteOn vendor IE fingerprints (tag 221 / OUI 50:6F:9A).

Effectiveness / confidence

Highest among stock methods (confidence ~100 for wildcard + OUI; high 80s for general probe + OUI). Field reports from DeFlockJoplin-style testing describe strong capture rates when cameras are in uplink-hunt behavior, with far fewer false positives than OUI-only matching.

When it works best

- Camera is actively enumerating networks (STA uplink discovery).
- Channel plan covers the hop sequence (Primary or Full/Asc as needed).
- Dwell time is short enough to intersect rapid probe bursts (~280 ms default).

Limitations

- No emission → no detection, regardless of method quality.
- Non-Flock devices can also send wildcard probes (OUI gate is essential).
- Requires being on the right channel during the burst.

Best practices

- Leave Wildcard Probe **ON** for virtually all field work.
- Prefer Primary 1/6/11 for speed; switch to Full if you suspect off-channel activity.
- Treat WILD PROBE hits as high priority for mapping and verification.

6.4 BLE Scanning (Secondary)

How it works

When enabled, the firmware periodically pauses WiFi completely, brings up NimBLE, runs a short blocking scan (~2 s), parses advertisements for manufacturer ID **0x09C8 (XUNTONG)**, Flock-related OUIs, name keywords, and Raven-related service UUID bases, then tears down BLE and restarts WiFi promiscuous mode. WiFi-only slices of ~12 s separate BLE windows so the primary detector remains dominant.

Effectiveness / confidence

Variable and often **lower than WiFi probe methods** on current deployments. Useful as a secondary net for devices that still advertise distinctive BLE data, but community reports indicate BLE has become less reliable after vendor changes.

When it works best

- Target hardware still emits XUNTONG manufacturer data or Raven UUID patterns.
- Stationary or slow surveys where a 2-second WiFi gap is acceptable.

Limitations

- Classic ESP32 cannot sniff WiFi and BLE at the same instant.
- UI may freeze briefly during BLE slices.
- Init cost and memory pressure on no-PSRAM CYD hardware.

Best practices

- Leave BLE **OFF** for high-speed driving surveys focused on WiFi probes.
- Enable BLE for walking / soak tests when hunting older or Raven-class signatures.
- If stability problems appear, disable BLE; WiFi remains the primary detector.

6.5 Additional Supporting Paths

Path	Default	Role
SSID keywords	ON	Matches fragments such as flock, flock, raven in probe/beacon SSIDs
Broad OUI	OFF	Any frame from known OUI — noisy; experts only
LiteOn vendor IE	automatic	Raises confidence when probe carries IE OUI 50:6F:9A

6.6 Channel Hopping Plans

Plan	Channels	Dwell (default)	Best for
Primary	1, 6, 11	280 ms	Driving; fast cover of busy spectrum
Full	1–13	280 ms	Maximum catch; slower revisit rate
Ascending	1–11	280 ms	Align with observed ascending probe hops

7. Alerts & Sensitivity

7.1 Settings Overview

Setting	What it does
Audio (buzzer)	Plays a multi-tone pattern on de-duplicated hits (GPIO 22 piezo)
RGB LED alert	Flashes red on hit; soft blue pulse while scanning; green when idle
SD logging	Appends CSV rows when a FAT32 card is mounted (see Section 8)
Channel plan	Primary / Full / Asc — changes hop set immediately

7.2 De-duplication / Cooldown

Raw RF can deliver dozens of identical frames per second from one device. The firmware keeps a small per-MAC cooldown table (default **4 seconds**). Repeat frames from the same MAC within the window still update statistics as appropriate, but they do **not** re-trigger full-screen alert, buzzer, or LED spam. This keeps the device usable in a vehicle without constant alarms when parked near a talkative unit.

7.3 RSSI Floor

Frames weaker than the configured floor (default **−95 dBm**) are ignored in the sniffer path. Raising the floor (for example to **−80 dBm**) reduces distant noise at the cost of range; lowering it increases sensitivity and multipath clutter. Adjust in `include/config.h` (`RSSI_FLOOR`) for advanced tuning.

7.4 Recommended Profiles

Scenario	Methods	Channel plan	BLE	Audio	Notes
Driving (default)	OUI TX/RX + Wildcard + SSID	Primary 1/6/11	Off	On	Maximize WiFi time; short dwell
Walking survey	Same as driving	Primary or Full	Optional	On	Slower speed → better catch
Stationary soak	All WiFi methods	Full or Asc	Optional	Optional	Log to SD; longer observation
Low false positives	Wildcard + OUI TX; RX optional	Primary	Off	On	Broad OUI off; triage by conf.
Aggressive research	All + Broad OUI	Full	On	On	Expect noise; verify manually

7.5 Reducing False Positives

- Keep **Broad OUI** disabled unless you are actively researching.
- Prefer hits with method **WILD PROBE** / high confidence scores.
- Require geographic sense: a single weak OUI RX hit mid-highway may be multipath or coincidence.
- Use SD logs later to cluster by MAC and time rather than reacting to every beep.
- Update OUIs carefully — adding overly broad prefixes increases noise.

8. Logging & microSD Card

8.1 Enabling Logging

At boot the firmware probes the onboard microSD slot. If a readable card is found, logging is turned **ON** automatically and the Home screen shows **SD:ready**. You can also toggle **SD logging** under **Alerts & Sensitivity**. Turning it on re-probes the slot (hot-plug friendly).

Format requirement

The microSD card **must be formatted as FAT32**. exFAT/NTFS are not supported by the Arduino SD stack used here. Cards that are empty or newly purchased should be formatted FAT32 on a computer before first use.

8.2 Log Location and Format

File path on the card: **/gfy_log.csv** (appears as gfy_log.csv in the card root on a desktop OS).

Header row:

```
millis,uptime_s,mac,oui,method,protocol,rssi,channel,confidence,label
```

Column	Description
millis	Device uptime timestamp in milliseconds at write
uptime_s	Same time in whole seconds (handy for spreadsheets)
mac	Full six-octet MAC (colon-separated hex)
oui	First three octets only
method	Human method name (e.g. WILD PROBE, OUI RX)
protocol	wifi or ble
rssi	Signal strength in dBm
channel	2.4 GHz channel of the capture
confidence	0–100 score from method ranking
label	SSID snippet, <wildcard>, BLE name, etc. (commas stripped)

8.3 Retrieving Logs

- Stop scanning if desired, power down, and remove the microSD card.
- Open gfy_log.csv in any spreadsheet or text editor.
- Sort by confidence, filter method == WILD PROBE, or pivot by OUI for analysis.
- Combine with GPS tracks from a separate phone app using approximate timestamps (the device log is relative to boot millis, not wall clock unless you note start time).

8.4 Tips for Reliable Logging

- Use reputable name-brand cards; tiny/no-name cards fail more often on SPI buses.
- Eject safely from the OS after copying; avoid pulling the card during a write.
- If Home shows **SD:none**, reseal the card and toggle SD logging or reboot.

- Serial will print [log] SD wrote hit #N on success.
- Touch and SD share SPI hardware resources on the CYD; the firmware re-claims the bus around each write — still, avoid very rapid forced probes while touching the UI.

9. Best Practices & Tips

9.1 Mounting and Antenna Considerations

- The ESP32-WROOM module antenna is PCB/metal-can based — avoid covering it with hands, metal plates, or dense carbon dash mats.
- Orient the board so the module antenna faces outward (toward glass), not into a metal console.
- Do not enclose the CYD in a fully metal box without an external antenna modification (advanced; not covered by stock firmware).
- Keep USB power leads short and well-seated to prevent brownout resets.

9.2 Driving vs Walking Strategies

Mode	Strategy
Driving	Primary plan 1/6/11; BLE off; audio on; watch for clusters of high-confidence hits near poles
Driving (dense urban)	Slightly lower speed on first pass; consider Full plan on a second loop
Walking	Full or Asc plan; optional BLE; pause near suspected poles for 30–60 s
Stationary	Full plan + SD logging; note wall-clock start time for later correlation

9.3 Interpreting Results

- **WILD PROBE + strong RSSI** — Highest priority; treat as likely nearby ALPR-class radio.
- **PROBE OUI** — Strong candidate; confirm with geography and repeats.
- **OUI RX only** — Indirect; look for accompanying TX/probe evidence.
- **Broad OUI / weak RSSI one-offs** — Log for research; do not over-claim.
- **Frames rising, hits zero** — Radio is healthy; no signature match yet (common and OK).

9.4 Updating the OUI List

OUIs live in `include/oui_list.h` as a static array of three-byte prefixes with a confidence tier. To update:

- Add new high-confidence prefixes only from documented research or your own verified captures.
- Keep lower-confidence prefixes marked with higher conf tier values so scoring can demote them.
- Rebuild and flash: `pio run -e cyd -t upload`.
- Record why each OUI was added (source, date) in your research notes — future firmware churn will otherwise make the list opaque.

9.5 Operational Safety

- Do not operate a touchscreen or react to alerts in a way that distracts from driving.
- Prefer a passenger operator or review SD logs after the route.
- Obey traffic laws; this tool is for observation, not confrontation.

10. Credits & References

10.1 Project

GoFlockYourself firmware and CYD interface — @OffGridPete and contributors. Version documented: **1.0.0** (12 August 2026).

10.2 Research and Prior Art

Contributor	Contribution
@NitekryDPaul	Core WiFi OUI corpus; addr1 receiver-side detection technique
Michael / DeFlockJoplin	Wildcard probe signature; field validation; OUI 82:6B:F2
colonelpnichacks / flock-you	Promiscuous detection pipeline reference implementation
GainSec	Raven BLE service UUID research
DeFlock community	Crowdsourced ALPR mapping methodologies and public datasets
IEEE registration	MA-L B4:1E:52 allocated to Flock Safety (2024-05-09)
Pintor & Atzori (2022)	Academic work on probe-request IE fingerprinting (context for IE methods)

10.3 Software Stack

- Espressif Arduino-ESP32 core and WiFi promiscuous APIs
- Bodmer TFT_eSPI — ILI9341 display driver
- Paul Stoffregen XPT2046_Touchscreen — resistive touch
- h2zero NimBLE-Arduino — optional BLE scanning
- PlatformIO — build and flash workflow

10.4 Disclaimer

This documentation and firmware are provided for security research, privacy auditing, journalism, and education. Authors and contributors assume no liability for misuse. The reception of broadcast 802.11/BLE advertisements does not include authorization to access private networks, tamper with infrastructure, or violate local regulations. Always verify legal context in your jurisdiction.

End of main documentation — appendices follow.

Appendix A — Confidence Scoring

Each detection method maps to a base confidence score used in the UI and CSV log. Lower-confidence OUI table entries slightly demote the score at enqueue time.

Method	Base score	Notes
WILD PROBE	100	Known OUI + empty SSID probe
LITEON IE	95	Vendor IE 50:6F:9A on probe
PROBE OUI	85	Any probe request from known OUI
SSID KEY	80	Keyword match in SSID
BLE UUID	75	Raven-class service UUID
BLE MFG	70	XUNTONG manufacturer ID 0x09C8
BLE NAME	65	Name keyword match
OUI TX	60	Transmitter OUI (non-probe / broad paths)
BLE OUI	55	BLE address OUI only
OUI RX	50	Receiver addr1 OUI

Appendix B — Serial Diagnostics

USB serial runs at **115200** baud. Useful patterns:

```
[wifi] SCAN start plan=0 dwell=280ms ch=1
[hb] ch=6 any=12040 mgmt=8900 probe=210 wild=12 oui=3 hits=1/1 heap=145000 scan=1
[HIT] WILD PROBE AA:BB:CC:DD:EE:FF ch=6 rssi=-62 conf=100 WiFi <wildcard>
[log] SD wrote hit #3 → /gfy_log.csv
```

Observation	Interpretation
any stays 0	Radio not receiving — check init, antenna, environment
any > 0, hits = 0	Healthy RF; no Flock signatures yet
heap falling hard	Investigate BLE on/off; reboot if unstable
SD re-mount failed	Reseat card; confirm FAT32; try another card

Appendix C — Source File Map

Path	Role
platformio.ini	CYD environment, TFT_eSPI pins, libraries
partitions.csv	Flash layout (app + SPIFFS region)
include/config.h	Pins, defaults, timing constants
include/oui_list.h	Compiled OUI table and keyword lists
include/types.h	Events, settings, method names/scores
src/main.cpp	setup/loop orchestration
src/radio.cpp	Promiscuous sniffer, hop, BLE handoff
src/detection.cpp	Queue, cooldown, history
src/ui.cpp	Touch UI screens and alerts
src/logger.cpp	Serial + microSD CSV logging
src/hardware.cpp	RGB LED and piezo helpers
README.md	Quick-start companion to this document

Document control

Title: GoFlockYourself Technical Documentation · Version 1.0.0 · Date: 12 August 2026 · Audience: makers, privacy researchers, and field auditors · Classification: public research aid.